Phase 2: Data Collection, Understanding and Preparation

Project: Few-Shot Colorectal Histology Classification

Group ID: F26-04 | GitHub: https://github.com/Wajiha-144/Few-shot-colorectal-histology

1. Dataset Source and Description

The project uses a single dataset: Colorectal Histology MNIST on Kaggle (kmader/colorectal-histology-mnist), which mirrors the Kather et al. (2016) colorectal cancer texture dataset, "Multi-class texture analysis in colorectal cancer histology" (Scientific Reports, 6:27988). No other dataset is used.

The Kaggle page provides several versions of the same images. We use the folder of full-colour 150×150 pixel RGB tiles (5,000 images in 8 class folders). The 28×28 and 64×64 CSV versions are not used, because downsampling removes colour and cellular detail and the Phase 1 plan requires 224×224 input for a pretrained vision transformer.

The tiles are haematoxylin-and-eosin (H&E) stained human colorectal tissue patches. The dataset has 5,000 images in 8 tissue classes with 625 images per class.

Label	Tissue class	Folder name	Images
1	Tumour epithelium	01_TUMOR	625
2	Simple stroma	02_STROMA	625
3	Complex stroma	03_COMPLEX	625
4	Immune cell conglomerates (lympho)	04_LYMPHO	625
5	Debris and mucus	05_DEBRIS	625
6	Normal mucosal glands	06_MUCOSA	625
7	Adipose tissue	07_ADIPOSE	625
8	Background (empty)	08_EMPTY	625

Table 1. Class labels, folder names and image counts (the notebook checks the folder names against this table).

2. Data Collection Method

This is secondary data. No new images were collected. Kather et al. produced the images from anonymised H&E-stained colorectal cancer histology slides, which were digitised, cut into tiles and labelled by tissue type. We downloaded the dataset from Kaggle.

Acquisition is automated in the notebook (notebooks/F26-04.ipynb):

1. The notebook looks for the tile folders in the path set by the KATHER_DATA environment variable, in data/raw, or in the Kaggle input folder.
2. If they are not found, it downloads the dataset with the kagglehub library (this needs Kaggle credentials). Manual download from the Kaggle page also works.
3. The raw data is never committed to the repository (data/ is listed in .gitignore). A requirements.txt lists the libraries used.

3. Cleaning and Preprocessing Steps

3.1 Cleaning checks

Each step is performed and logged by the notebook (results/cleaning_report.json):

- **Index and label verification.** An index of all images is built, and each label (1 to 8) is taken from the folder number and checked against the expected class folders.
- **File integrity.** Every file is opened and verified; unreadable files would be dropped. Corrupt files found: 0.
- **Size and colour mode.** All tiles are checked to be 150×150 RGB. Sizes found: 150x150: 5000; modes found: 5000. Any tile of a different size would be resized to 150×150 and counted.
- **Missing values.** Missing values in the index: 0.
- **Duplicates.** Exact duplicate images are detected by hashing pixel data, and the first copy is kept. Duplicates removed: 0.
- **Final count.** Images after cleaning: 5,000, with 625 per class. Pixel values range from 0 to 255 (8-bit).

3.2 Preprocessing for modelling

Images are stored as 150×150 uint8 RGB arrays. They are not converted to grayscale or flattened. The following are applied on the fly in the modelling phase, using functions defined in the notebook:

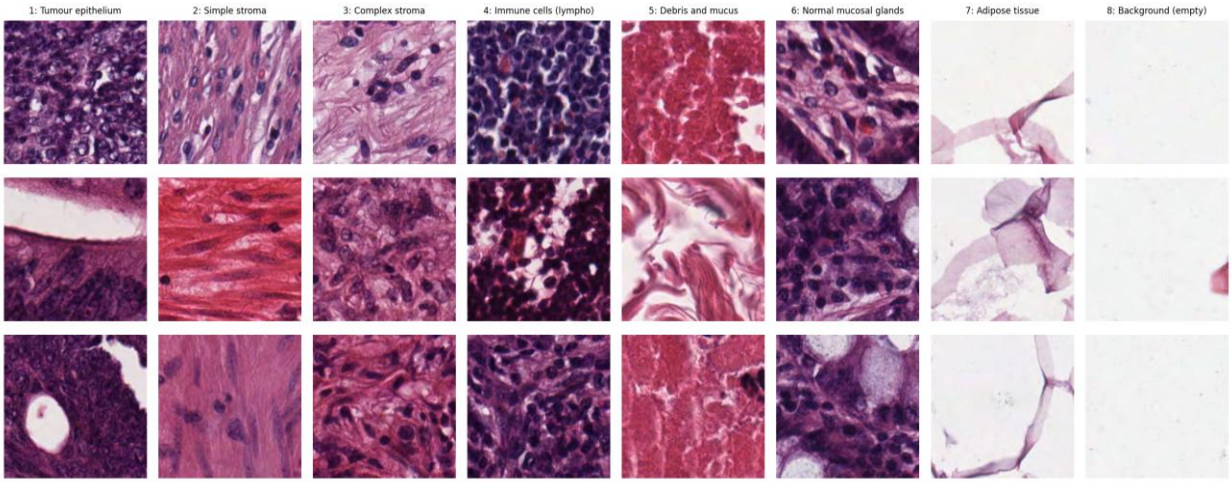
- **Resize and normalise:** bicubic resize to 224×224, scaling to [0,1], then ImageNet mean/std normalisation, giving a channels-first float32 array for the ViT and CNN backbones.
- **Training augmentation:** random flips, 90° rotations and mild colour jitter, applied only to training episodes.
- **Synthetic atypical samples:** strong stain/colour shift, heavy blur, or a patch pasted from another image. These are used in Phase 3 by the consistency loss of the proposed method.

Validation and test images only receive the resize and normalisation step.

4. Exploratory Data Analysis

The notebook produces the following analyses (saved in figures/ and results/):

- Example tiles for every class, and a class-balance chart.
- Per-class colour and texture statistics: brightness, contrast, mean R/G/B and gradient energy (a simple texture measure).
- A descriptive separability check: a nearest-centroid classifier on simple colour and texture features, with a confusion matrix and a class-centroid distance heatmap. This shows which classes look alike. It is a data description, not a model result.



Key insights from the exploration:

- **Perfectly balanced classes.** Every class has 625 images (12.5% each), so there is no class imbalance to correct. This is useful for few-shot work because any class can serve as a support or query class without sampling bias, and 625 images per class comfortably covers 1-, 5- and 10-shot support sets plus query images.
- **Clean data.** There are no missing values and the pixel range is valid, so the dataset is usable without cleaning.
- **Visually distinct brightness and texture patterns.** In the examples shown, classes 7 (adipose) and 8 (background) are bright with little texture, class 6 shows rounded gland-like structures, and class 4 shows high-contrast dark and bright regions. Classes 1, 3 and 5 appear darker and more fine-grained, so we expect these to be harder to separate than the bright classes. This is an observation from one image per class and should be confirmed with per-class pixel statistics.
- **Low resolution.** At 28 x 28 pixels, the fine nuclear and cellular detail visible in the original 150 x 150 tiles is lost, and only coarse tone and texture remain. This is a limitation to keep in mind when interpreting results.

5. Class Splits and Episode Protocol

Because only one dataset is used, the cross-domain protocol in the Phase 1 plan is not possible. Eight classes also cannot support 5-way few-shot with disjoint base and novel classes (a 5-way test would leave only 3 base classes). The protocol is therefore in-domain: 4 base classes, 4 novel classes, and 4-way 1-, 5- and 10-shot episodes.

Split	Base classes (training)	Novel classes (testing)
A (primary)	2 Simple stroma, 5 Debris, 7 Adipose, 8 Background	1 Tumour epithelium, 3 Complex stroma, 4 Lympho, 6 Normal mucosa
B (alternate)	1, 4, 6, 8	2, 3, 5, 7
C (alternate)	3, 4, 7, 8	1, 2, 5, 6

Table 3. Fixed class splits. They were set before any model was trained and will not be changed after seeing results.

- Split A puts the diagnostically meaningful tissue types in the novel set, and the visually easy classes (background, adipose) in the base set, so test accuracy is not inflated by trivial classes. Splits B and C are used to check that conclusions do not depend on one partition.
- Base validation: with so few classes a separate validation class set is not possible, so a stratified 20% of each base class's images is held out for model selection. Novel-class images are never used for training or selection.
- Test episodes: 600 per shot setting per split, 15 query images per class. Episode i uses seed 10000 + i. The class set and query images are identical for 1-, 5- and 10-shot, and the support sets are nested (1-shot inside 5-shot inside 10-shot), so every method sees exactly the same tasks.
- Validation episodes: 300 per shot setting, drawn from the held-out base images (seed 20000 + i). Training episodes are sampled on the fly from base training images.
- The notebook runs automated checks that episodes are reproducible from their seeds, that support and query never overlap, and that no novel-class image appears in any base role. The checks must all pass.

6. Final Dataset Summary

Property	Value
Source	Colorectal Histology MNIST (Kaggle), from Kather et al. (2016)
Files used	150×150 RGB tile folders (CSV versions not used)
Images after cleaning	5,000 expected (see Section 3 for the actual count)
Classes	8, 625 images each, perfectly balanced
Stored format	uint8 RGB, 150×150
Model input	224×224 float32, ImageNet-normalised (on the fly)
Protocol	In-domain, 4-way, 1/5/10-shot, 15 queries per class
Splits	A (primary), B, C: 4 base and 4 novel classes each
Episodes	600 test and 300 validation per shot per split, with saved seeds and image IDs

Table 4. Final prepared dataset.

7. Reproducibility

Repository layout: data/raw (not committed); data/processed/index_clean.csv; splits/ (class_splits.json, image_roles.csv, test_episodes_{A,B,C}.json, val_episodes_{A,B,C}.json); results/ (cleaning report, EDA tables, summary); figures/; notebooks/01_phase2_data_preparation.ipynb; requirements.txt. Running the notebook from top to bottom regenerates every file, with a global seed of 42.

8. Limitations

- Single source: all images come from one dataset, so results measure generalisation to new tissue classes in the same domain, not to new scanners, stains or cancer sites.
- Only 8 classes, so only 4 novel classes per split. Using three splits reduces, but does not remove, dependence on the class partition.
- Patient or slide identifiers are not provided. Images from the same slide may appear in both support and query sets, or in both base training and base validation sets, which may make results optimistic.
- Tiles are small (150×150 pixels), so large-scale tissue architecture is not captured.

References

Kather, J. N., Weis, C.-A., Bianconi, F., et al. (2016). Multi-class texture analysis in colorectal cancer histology. Scientific Reports, 6, 27988.

Kaggle dataset: Colorectal Histology MNIST (kmader/colorectal-histology-mnist). https://www.kaggle.com/datasets/kmader/colorectal-histology-mnist

Shakeri, F., et al. (2022). FHIST: A Benchmark for Few-shot Classification of Histological Images. arXiv:2206.00092 (background reference).